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Accurate Decomposition of Full Waveform Sonic Data

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Abstract

Full waveform sonic data is typically dominated by strong direct wave mode propagating along the borehole, which obscures the weak reflected signals of interest. Therefore, accurate wave mode decomposition is an essential step in enhancing the interpretability of full waveform sonic data, particularly for applications such as velocity estimation and imaging of near-borehole structures. We propose a novel decomposition technique based on a 3D sparse Radon transform that effectively separates overlapping wave in full waveform sonic data. The proposed method first estimates the slowness of different dominant linear events in common-shot gather through semblance analysis and then compute the dip of direct wave mode in common-offset gather using structure tensor. After that, the full waveform sonic data, originally in the time-space domain, is transformed into a time–slowness–space domain using a 3D sparse Radon transform. This transformation creates a sparse, higher-dimensional representation in which different wave modes become more separable due to their distinct orientations and slowness characteristics. A mask function is then automatically constructed in the Radon domain by analyzing signal orientation in both common-shot and common-offset gathers. This enables the selection and isolation of specific wave mode components, including weak reflections and directionally distinct upgoing and downgoing reflected waves. Finally, an inverse transform reconstructs each wave component into the original time-space domain, yielding high-fidelity decomposed wavefields. Numerical tests on field monopole full waveform sonic datasets demonstrate that the proposed method significantly improves the signal-to-noise ratio (SNR) by suppressing random noise and enhancing coherent energy. Furthermore, it accurately decomposes the monopole dataset into six physically meaningful components: direct P-waves, direct S-waves, Stoneley waves, reflected waves, upgoing and downgoing reflections. This approach provides a powerful processing tool for acoustic logging interpretation and reflection imaging applications in complex subsurface environments.

Introduction

Acoustic reflection imaging has become a powerful tool for characterizing near-borehole geological structures, offering high-resolution insights into fractures, vugs, and thin formations that are otherwise undetectable through traditional sonic logging or surface seismic methods (Hornby, 1989; Tang, 2004; Zheng and Tang, 2005; Chen et al., 2019; Kong et al., 2024). This technique utilizes weak body wave

reflections to provide valuable structural information tens of meters into the formation, such as near-borehole fault analysis, adjacent borehole detection, and hydraulic fracturing evaluation etc. Despite its potential, effective application of acoustic reflection imaging remains limited by two major challenges: the overwhelming presence of direct waves and ground rolls traveling along the borehole, which obscure the reflection data necessary for imaging, and the inherent complexity of wavefields caused by overlapping events and formation heterogeneities.

The direct mode waves of borehole full waveform sonic data are confined in the fluid-filled borehole as P-, S-, and Stoneley-waves for a monopole source, or as tube waves for a dipole source, which greatly contaminates the reflection mode due to its dominant and dispersive energy. Consequently, extracting the reflected waves becomes a crucial pre-processing step and has attracted considerable attention from researchers. Moreover, separating the upgoing and downgoing reflections from the data and using them individually for imaging can remove ghost events caused by mixed reflection from the image result (Tang et al., 2007). Besides, although the direct mode waves are considered as noise for reflection imaging, those direct mode waves can be used for estimating velocity, anisotropy, attenuation, and lithology of the formation adjacent to the borehole (Tang, 2003; Tang and Cheng, 2004). Therefore, decomposing full waveform sonic data into different wave modes is essential.

Since Hornby (1989) introduced imaging of near-borehole structures using full-waveform sonic data, many techniques have emerged to extract reflected waves. Li et al. (2002) applied high-pass filter and median filter to remove the tube waves and direct arrivals, followed by FK filter to separate the upgoing and downgoing reflected waves. While the median filter is rapid and effective, it can produce discontinuous reflection events due to the direction inconsistency between the direct wave modes and median filter. Tang et al. (2007) propose a parametric prediction method to predict deterministic direct waves using the given velocities and subtract them from total data, but accurate velocity model can be often difficult to estimate. Tao et al. (2008) present multi-scale semblance method for the separation of the reflected waves and direct waves in borehole acoustic logging. Li et al. (2017) propose using the Karhunen-Loève transform to decompose the full waveforms based on energy concentration, allowing direct waveforms to be reconstructed from dominant components while preserving residual reflections. Chen et al., (2019) use Shearlet transform and leverage hard thresholding in the Shearlet domain to isolate reflected waves. Kong et al. (2024) employ variation mode decomposition and curvelet transform to extract reflected waves from full waveform sonic data. While these methods demonstrate encouraging results, they are generally operated by 2D processing of common-offset data based on the different moveout characteristics between reflections and direct mode waves and often underutilize information embedded in high dimension.

To overcome these limitations, we propose a novel wave mode decomposition method based on the three-dimensional sparse Radon transform, which provides a robust means of separating overlapping wave modes in full waveform sonic data. Our method takes advantage of the fact that different wave modes propagate with distinct slowness in common-shot gather and directional characteristics in common-offset gather and therefore become more separable when represented in a higher-dimensional Radon domain.

In contrast to conventional methods, our 3D approach unifies information across spatial dimensions, allowing for improved resolution of overlapping events and more robust decomposition. We validate the proposed method using monopole full waveform sonic dataset. The results show that our method not only improves signal-to-noise ratio (SNR) by suppressing random noise and preserving coherent energy, but also successfully decomposes the monopole data into six meaningful components: direct P-waves, S-waves, Stoneley waves, reflected waves, upgoing, and downgoing reflections.

Method

The basic idea behind wave decomposition is to separate full waveform sonic data into independent components in Radon domain and automatically identify them. The proposed workflow consists of four

main steps. First, we estimate the slowness of dominant direct wave modes in common-shot gathers through semblance measurement and compute the orientation of direct waves in common-offset gather through structure tensor method. Second, the original full waveform sonic data is transformed into a 3D sparse Radon domain, enhancing the sparsity and focusing of different wave modes. Third, a mask function based on energy orientation in both gathers is applied to isolate desired components. Finally, an inverse transformation restores each selected component to the original domain, enabling separate analysis or imaging.

3D Sparse Radon transform

We propose to use a 3D sparse Radon transform for different wave decomposition and enhancement of full waveform sonic data. Specifically, the proposed method can decompose the data acquired using monopole array sonic into 6 components including P-waves, S-waves, Stoneley waves, reflections waves, upgoing and downgoing reflections, and decompose the dipole full waveform sonic data into 4 components including tube waves, reflections waves, upgoing and downgoing reflections. Besides, the proposed method can recover sparse signals with sufficient signal-to-noise ratio (SNR).

The separated component $d(t, x_r, x_s)$ can be obtained using the Radon transform equation below,

$$d(t, x_r, x_s) = \iiint M(\tau, p_s, p_r, x') u(\tau, p_s, p_r, x') dp_r dp_s dx' + e(t, x_r, x_s), \quad (1)$$

$$\tau = t - p_s(x_s + x') - p_r x_r \quad (2)$$

where p_s and p_r stand for the slowness parameter along source and receiver dimension, respectively, τ is the intercept, $e(t, x_r, x_s)$ represents the random noise and unmodeled scattered waveforms, $u(\tau, p_s, p_r, x')$ represents the 4D volume of sparse Radon-domain signals, x' is the center along shot dimension for a given 3D window, and t is the time. The length of the window along the shot dimension should ensure the signal within the window can be approximated as linear event. Given that full waveform sonic data typically have a small number of receivers in practice, the signal along the receiver dimension is already approximated to be linear, therefore, we split the 3D window along the shot dimension. If the number of receivers is large so that the event linearity along receiver dimension does not hold, we need to split both the shot and receiver dimensions to ensure data linearity within one 3D window.

For monopole logging data, $d(t, x_r, x_s)$ can be P-, S-, Stoneley-waves, reflections, upgoing or downgoing reflections, depending on the mask function $M(\tau, p_s, p_r, x')$ provided. By designing a specific mask function based on the behavior of the required wave mode, we can isolate the corresponding component in the radon domain. Following an inverse radon transform, we can obtain the required isolated waves in time-space domain. Therefore, constructing the 4D sparse volume $u(\tau, p_s, p_r, x')$ in Radon domain is crucial for separating different waves.

To construct the sparse Radon-domain signals $\mathbf{u}$, we formulate the objective function as

$$\min \|\mathbf{L}\mathbf{u} - \mathbf{d}\|_2^2 + \lambda \|\mathbf{u}\|_1, \quad (3)$$

where $\mathbf{L}$ is the 3D Radon operator formulated in [equation 1](#), $\mathbf{d}$ is the input data from one 3D window, $\mathbf{u}$ represents the Radon-domain signals, and λ is the regularization parameter. We apply an iterative soft thresholding method to solve the optimization problem defined in [equation 3](#) that leads to an l_1 -norm minimization of the model. This step aims to estimate the sparsity level of the solution and provide a good initial estimate for the next stage. The iterative soft thresholding method updates the Radon-domain signals by

$$\mathbf{u}_{k+1} = \mathbf{T}(\mathbf{u}_k + \alpha \mathbf{L}^T \mathbf{L} \mathbf{u}_k - \alpha \mathbf{L}^T \mathbf{d}), \quad (4)$$

where $\mathbf{L}^T$ is the adjoint Radon transform that transforms time-space domain to the Radon domain, α is the step length. $\mathbf{T}$ is the shrinkage threshold operator defined as,

$$\mathbf{T} = \text{sign}(\mathbf{u}) \max\{|\mathbf{u}| - \mathbf{C}(\mathbf{m})\lambda, 0\}, \quad (5)$$

where $\text{sign}(\mathbf{u})$ indicates the sign of the component, and operator $\mathbf{C}$ computes the average value of absolute $\mathbf{m}$. At each iteration, $\mathbf{L}^T \mathbf{L} \mathbf{u}_k - \mathbf{L}^T \mathbf{d}$ resembles a gradient for the l_2 -norm term in the objective function, and $\mathbf{T}$ enhances sparsity in Radon domain by reducing the magnitude of each nonzero component. It is proved that the shrinkage-thresholding iteration provides the optimal solution to the l_1 -norm regularized minimization problem within a finite number of iterations, given appropriate conditions (Hale et al., 2008). Once an approximate solution $\mathbf{m}$ of the l_1 -norm regularized minimization problem is obtained, we can isolate the required wavefield by designing the mask function M to include the desired wave mode before transforming the data back to the time-space domain.

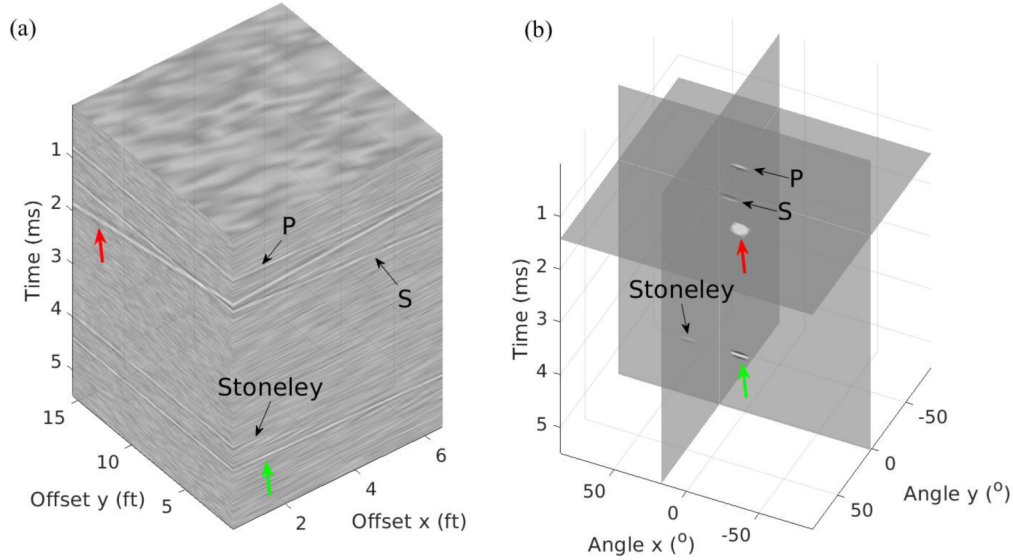


Figure 1—A synthetic example including (a) input noisy data and (b) result in sparse Radon domain.

Semblance analysis

To design the mask function for wave separation, we need first identify the dips of the direct wave modes in common-shot ($x_s - t$) and common-offset ($x_r - t$) domain. In common-shot domain, we obtain the slowness of P-waves, S-waves, Stoneley waves by computing the stacking energy along different slowness values, i.e. computing the semblance measurement $E(s, x_s)$ (Kimball and Marzetta, 1984) using the equation,

$$E(s, x_s) = \frac{\sum_{l=-L/2}^{L/2} \left[\sum_{j=1}^N d(t = sx_j + l\Delta l, x_j, x_s) \right]^2}{N \sum_{l=-L/2}^{L/2} \sum_{j=1}^N d^2(t = sx_j + l\Delta l, x_j, x_s)}, \quad (6)$$

where s is the scanned slowness and L is the time window length. We then stack the semblance measurement $E(s, x_s)$ within one 3D window along the shot dimension, and the result can indicate the slowness of P-waves, S-waves, Stoneley waves for this given 3D window.

Based on the slowness of each type of wave and the interception, we can define the corresponding isolation range in the $p_r - \tau$ section. In the $p_s - \tau$ section, the P-waves, S-waves, Stoneley waves correspond to small angles, computed by structure tensor. Such criterion determines the 3D mask functions for isolating P-waves, S-waves, Stoneley waves.

The reflection waves are then separated by removing the components of P-waves, S-waves, Stoneley waves from the monopole logging data in sparse radon domain. The resulting reflection component in Radon domain can be further used to isolate the upgoing and downgoing reflected waves by applying an inverse transform only to the signals of negative or positive angles in $p_s - \tau$ sections, respectively.

Results

We validate the method on a monopole dataset from the Middle East. The data has 8987 shots with a shot interval of 0.5 ft and the nearest offset between the shot and the receiver is 10.7 ft. Time sampling is 0.01 ms with a time recording length 5.2 ms. Each source has 13 receivers with a receiver space interval of 0.5 ft. We first apply an amplitude balancing and band pass filter to the raw data. The corresponding results in Figure 2 indicate that the shear waves and Stoneley waves are dominant which overwhelm the reflections as well as background noise.

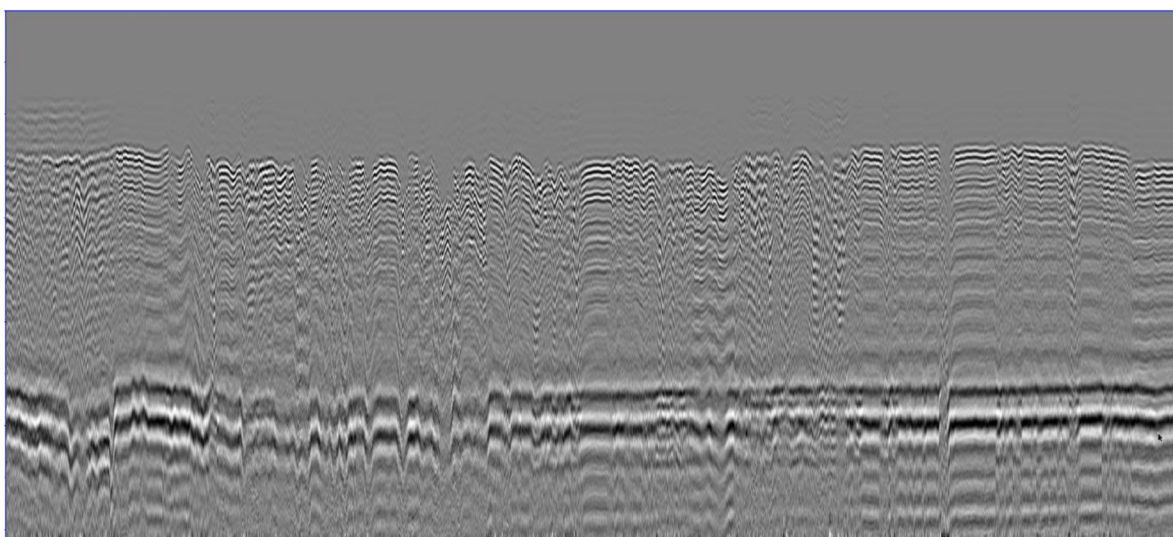


Figure 2—Monopole full waveform sonic data in one common-offset gather after amplitude balancing and bandpass filter.

To separate different waves, we then compute the semblance measurement of the monopole logging data for all the common-shot gathers and the result in Figure 3 shows the stacking energy of each common-shot gather corresponding to the different slowness. We then manually pick the slowness of P-, S-, and Stoneley-waves.

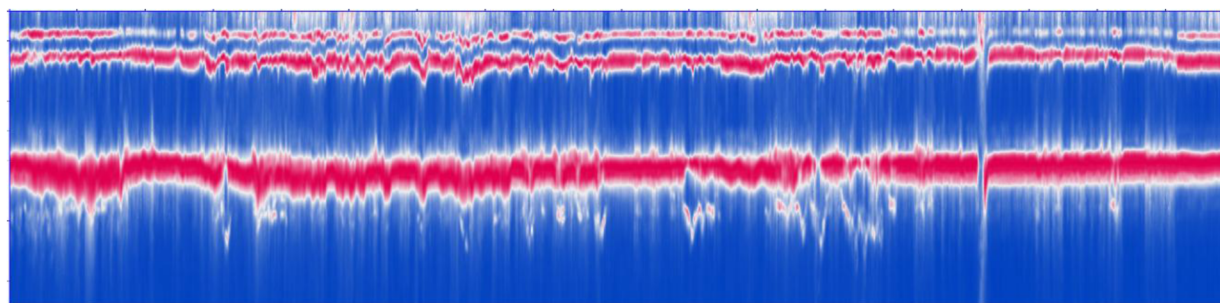


Figure 3—Semblance measurement of the monopole full waveform sonic data.

For a given 3D volume in a 3D window, we first identify the average slowness of P-waves, S-waves, Stoneley waves by stacking the semblance measurement along the shot dimension. The stacking range along the shot dimension matches the 3D window length along the shot dimension. With the identified slowness and the interceptions computed using the offset of the reference trace, we can design the mask function which can be further used to isolate the different wavefields. The window size along the shot dimension is set to 15 ft and the regularization parameter λ is 0.01 for the sparse Radon transform. By applying the sparse Radon transform and transforming all the components back to time-space domain, we can improve the S/N of the full waveform sonic data by enhancing the coherent signal and suppressing random noise.

The results in Figure 4 demonstrate that data quality is clearly improved compared with the input data in Figure 2. The striped noise and background noise are well suppressed, and the energy is well balanced from shallow to deep compared with the input data.

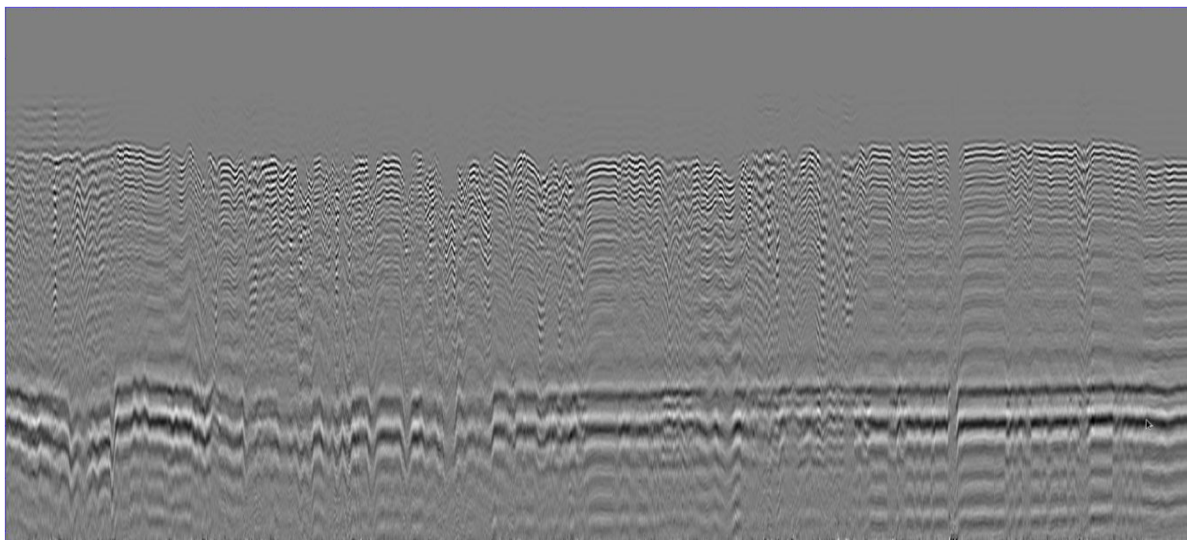


Figure 4—Denoised results in one common-offset gather.

Figures 5-10 are the decomposed P-waves, S-waves, and Stoneley waves, reflections, upgoing and downgoing reflection waves, respectively.



Figure 5—Decomposed P-waves in one common-offset gather.

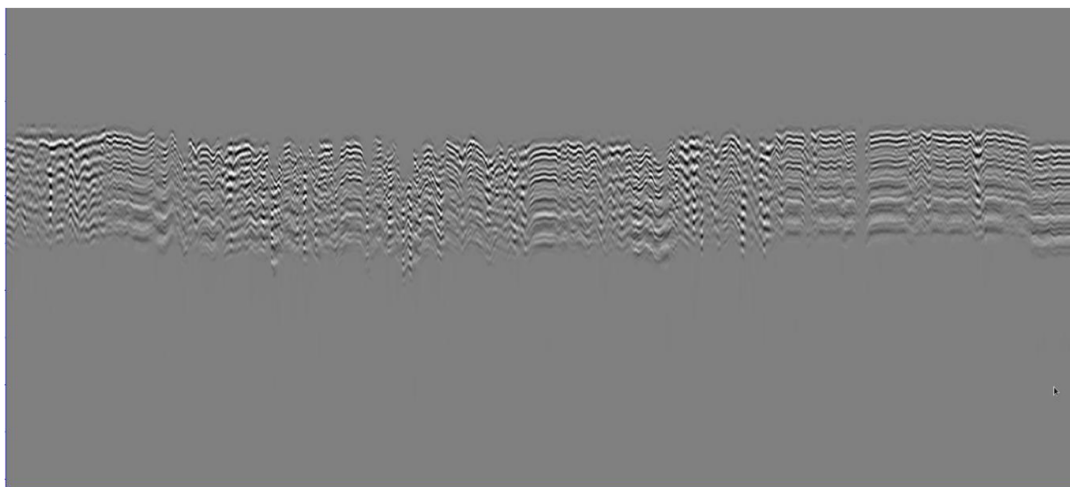


Figure 6—Decomposed S-waves in one common-offset gather.

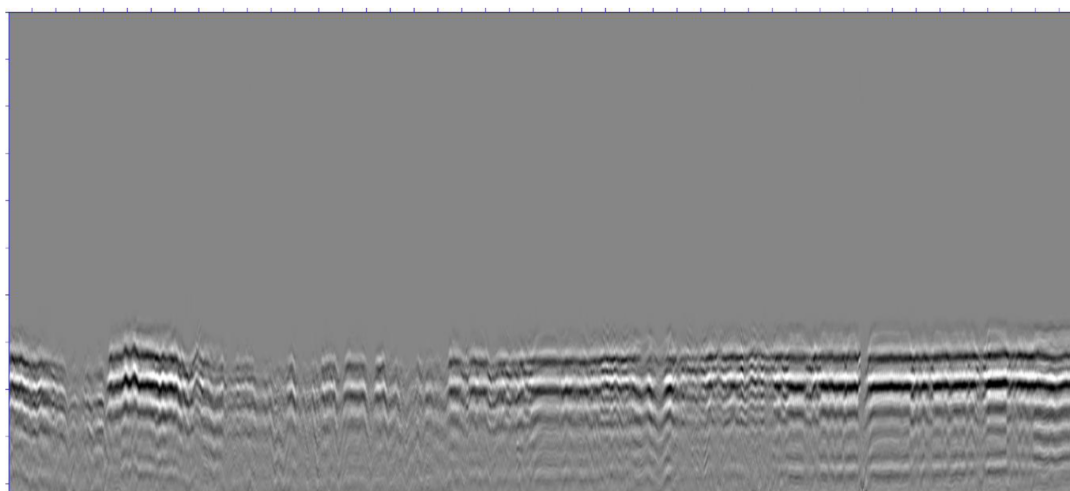


Figure 7—Decomposed Stoneley waves in one common-offset gather.

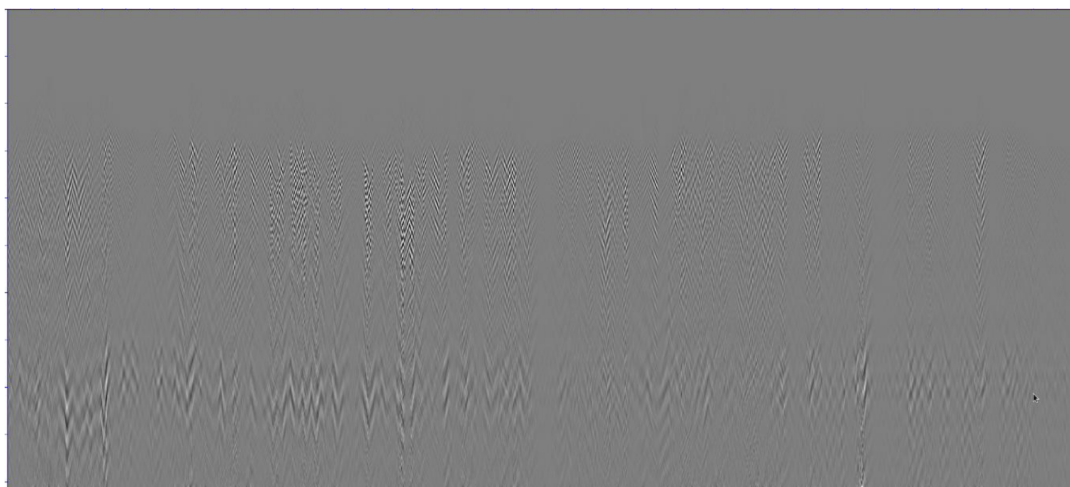


Figure 8—Decomposed reflection waves in one common-offset gather.

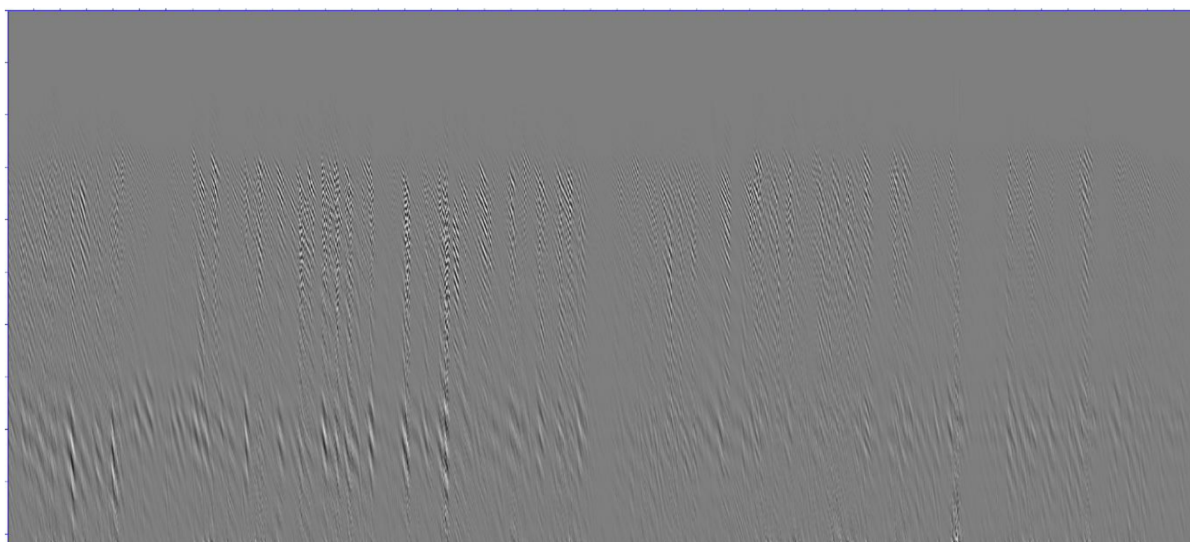


Figure 9—Decomposed upgoing reflection waves in one common-offset gather.

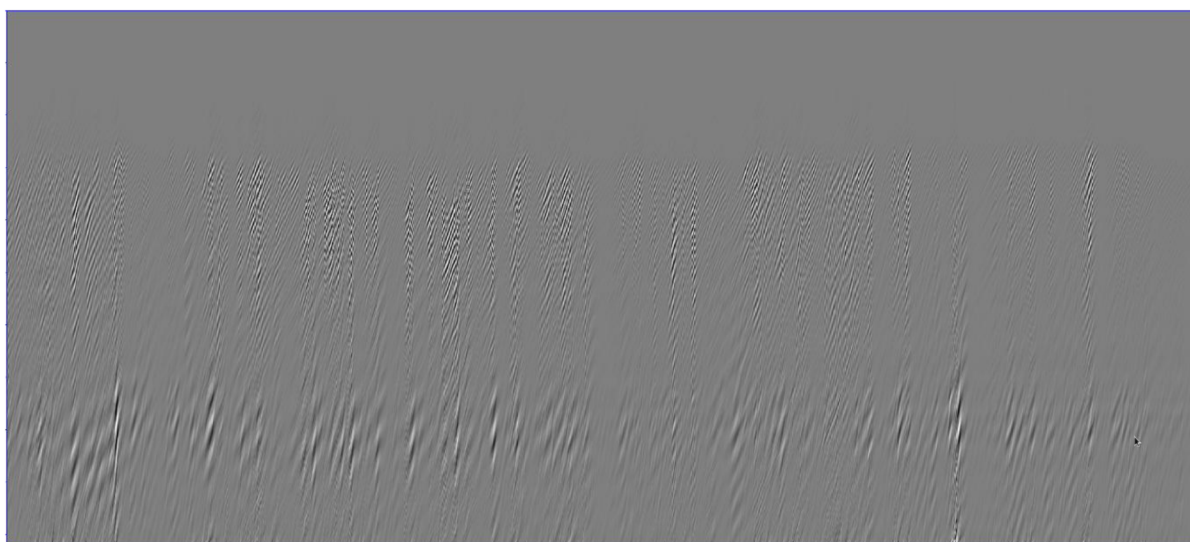


Figure 10—Decomposed downgoing reflection waves in one common-offset gather.

Conclusion

We present a new wave mode decomposition approach to decompose the full waveform sonic data into distinct components. This approach first transforms the 3D full waveform sonic data from time-space domain to time-slowness-space domain using 3D sparse radon transform, and then select the components corresponding to the required wave mode and transform the selected components back to time-space domain. Besides, the full waveform sonic data achieves high signal-to-noise ratio after sparse radon transform. Tests on the high-frequency far-offset monopole acoustic data validate that our method can greatly enhance the data quality and is effective for separating different wave modes including direct compressional waves, shear waves, Stoneley waves, reflection waves, upgoing and downgoing reflection waves.

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